# MECH0073 Experimental Rig and Prototype Handover

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## 1. Introduction

This document forms the primary handover record for the Aerosol Extraction System project, developed as part of MECH0073 MEng Capstone Group Design Project. The project aimed to computationally and experimentally investigate printzone aerodynamics in industrial inkjet printers, and to design, manufacture, and test an aerosol extraction system compatible with Agfa Inkjet Solutions' industrial printers. The project consisted of two hardware deliverables:

1. **Experimental Rig** — A reduced-scale lab rig for validating computational findings via shadowgraphy.
2. **Aerosol Extraction System Prototype** — A prototype extractor module designed for integration into Agfa industrial inkjet printers.

The next use of the experimental rig is planned for MECH0085 MSc Individual Project (MSc Engineering with Finance, supervised by Prof. José Rafael Castrejón-Pita), in collaboration with a global agricultural technology company, to investigate the effects of Stokes number and droplet deviation from background flow. The prototype extractor module is also to be held by Prof. José Rafael Castrejón-Pita at the Manufacturing Futures Lab at Marshgate, UCL East for use in a future, follow up student project in collaboration with Agfa Inkjet Solutions.

## 2. Physical Items

### *2.1. Experimental Rig*

The following physical components constitute the experimental rig assembly. Full engineering drawings and a detailed Bill of Materials are provided in Appendix C and Appendix B1 of the MECH0073 final report, respectively, and also at our central handover GitHub repository (a QR code is also available at the end of the document). All items are stored in the Manufacturing Futures Laboratory in Marshgate at UCL East under the care of Prof. José Rafael Castrejón-Pita.

### *2.2. Aerosol Extraction System Prototype*

The following physical components constitute the aerosol extraction system prototype. Full engineering drawings and a detailed Bill of Materials are provided in Appendix D and Appendix B2 of the Final Report, respectively and also at our central handover GitHub repository.

Table 1: Experimental Rig Bill of Materials

| **Item** | **Description** | **Qty** |
|---|---|---|
| Test Rig Experimental Setup | Full assembly | 1 |
| Test Rig Body | Sub-assembly | 1 |
| Test Rig Experimental Mist Capture | Sub-assembly | 1 |
| PLA 3-Way Profile Connector | 3D-printed frame connector | 2 |
| PLA Height Adj. Platform Rod Holder | 3D-printed platform support | 4 |
| Acrylic Height Adj. Platform Base | Adjustable platform | 1 |
| PLA Experimental Nozzle (S0–S4) | 3D-printed, five interchangeable geometries | 5 |
| PLA Vacuum Bleed | 3D-printed bleed body | 1 |
| PLA Test Rig Feet | 3D-printed | 4 |
| Acrylic Test Rig Base | Base plate | 1 |
| RS PRO Aluminium Profile 20×20 mm | Structural frame extrusion | 2 |
| RS PRO Angle Bracket 20×20 mm, 6 mm groove | Frame connector | 8 |
| Stainless Steel Needle Valve | F-to-F with male adaptors, suction control | 1 |
| Airbrush with Compressor (0.3 mm) | Aerosol generation unit | 1 |
| Conveyor Belt | Substrate drive unit | 1 |
| 163 cm Camera Tripod | Camera support | 1 |
| Differential Manometer | Testmeter ETI 9202 | 1 |
| Stainless Steel Shims 50×50×1 mm | Print-gap height setting | 10 |
| M4×10 Socket Cap Bolt | Standard fastener (ISO 4762) | 20 |
| M4 Flat Washer | Standard fastener (ISO 7089) | 8 |
| M4 Hex Nut | Standard fastener (ISO 4032) | 16 |

Table 2: Aerosol Extraction System Prototype Bill of Materials

| **Item** | **Description** | **Qty** |
|---|---|---|
| Prototype Extractor Module Body | Full assembly | 1 |
| Aluminium Extractor External Frame | Machined aluminium structural frame | 1 |
| PLA Extractor Shape Nozzle (S0–S4) | 3D-printed, five interchangeable inlet geometries | 5 |
| PLA Locating Block | 3D-printed nozzle guide blocks | 2 |
| Aluminium Vacuum Pump Connector | Machined channel connector | 1 |
| PLA Electronics Housing | 3D-printed enclosure for electronics | 1 |
| Extractor Gasket | Polymer airtight seal between nozzle and frame | 1 |
| Filter Foam | Replaceable filter medium | 1 |
| Sensirion SDP810-500Pa Differential Pressure Sensor | Digital pressure monitoring; upstream and downstream of filter | 2 |
| OLED Display 77×25.2 mm | Sensor readout display | 1 |
| RS PRO Hookup Wire Roll (PVC, 100 m) | Electronics wiring | 1 |
| RS PRO Straight Through-Hole Pin Headers | Electronics connectors | 1 |
| DHT11 Temperature and Humidity Sensor | Ambient condition monitoring | 1 |
| Arduino Uno Rev3 | Microcontroller for sensor data acquisition and display | 1 |
| Flexible Tubing 10 mm OD | Vacuum ducting between extractor and pump | 1 |
| M5×12 Socket Cap Bolt | Standard fastener (ISO 4762) | 10 |

#### 2.2.1. *Unused Materials and Consumables*

Any unused materials from the Bills of Materials listed above are stored in the Manufacturing Futures Laboratory at Marshgate, UCL East. Spare items include, but are not limited to:

- Remaining Aluminium Profiles
- Spare OLED monitor
- Remaining flexible tubing
- Remaining RS PRO hookup wire
- Stainless steel shims

### 2.3. *Technical Documentation*

The following documentation is included in the handover package. All digital files are accessible via the central handover GitHub repository.

Table 3: Technical Documentation Index

| **Document** | **Format** | **Location / Reference** |
|---|---|---|
| Final Project Report | PDF | GitHub repository |
| Engineering Drawings — Experimental Rig | PDF | Appendix C of Final Report, GitHub repository |
| Engineering Drawings — Aerosol Extraction System | PDF | Appendix D of Final Report, GitHub repository |
| CAD Files — Experimental Rig | f3d | GitHub repository |
| CAD Files — Aerosol Extraction System | f3d | GitHub repository |
| Bill of Materials — Experimental Rig | PDF | Appendix B1 of Final Report, GitHub repository |
| Bill of Materials — Aerosol Extraction System | PDF | Appendix B2 of Final Report, GitHub repository |
| Arduino Sensor Source Code | `.ino` | GitHub repository |
| OpenFOAM CFD Source Code | Text/shell | GitHub repository (with its own handover document and documentation repository). |
| Risk Assessments | PDF | Appendix A of Final Report, OSHENS risk management system, GitHub repository |
| Experimental Results | jpg/mp4 | GitHub repository |

## 3. Operating Instructions

### 3.1. *Experimental Rig Setup*

1. Assemble the aluminium profile frame using the 3-way connectors and angle brackets; tighten M4 socket cap bolts finger-tight before final torque.
2. Install the acrylic base and height adjustable platform. Set the print gap height as desired using precision shims placed at multiple spanwise locations. Secure the acrylic

ceiling via bolts on the platform rods.
3. Mount the airbrush nozzle flush with the underside of the acrylic ceiling sheet, oriented perpendicular to the substrate, using the retort stand. Fix the nozzle at the print gap centreline.
4. Insert the desired extractor nozzle geometry into the dedicated slot in the downstream acrylic panel.
5. Connect the vacuum cleaner to the bleed body with needle valve in line to regulate suction pressure.
6. Connect the differential manometer to the pressure tap on the extractor body to monitor suction gauge pressure.
7. Start the conveyor belt and allow it to reach steady state before data acquisition. The conveyor is operated at the maximum drive setting throughout all tests.
8. Activate the airbrush compressor. The airbrush spray force should be approximately 21.58 mN, as measured previously via precision scale.
9. Illuminate the printzone using the high-power lighting positioned to maximise contrast for shadowgraphy.
10. Experiments can be recorded by mounting an available camera on the tripod.

**Safety:** Always turn off the conveyor belt and vacuum when adjusting the extractor or airbrush position. Wear ear protection during prolonged operation. Refer to the MechSpace Risk Assessment (Appendix A2) if uncertain about operational risks before first use.

### *3.2. Aerosol Extraction System Prototype — Assembly and Integration*

1. Slot the desired inlet nozzle piece (S1–S4) into the base of the aluminium frame using the two PLA locating blocks as guides, and bolt into the frame using M3×12 socket cap bolts.
2. Seat filter foam or other filter medium on the interface between the frame and the vacuum pump connector.
3. Bolt the aluminium vacuum pump connector onto the top face of the aluminium frame using M5×12 socket cap bolts.
4. Electronic sensor box comes pre-assembled, but can be easily unbolted to make modifications.
5. Attach the 10mm OD pneumatic tubing from the sensor box into the hole insert in the static fixed jaw, and the 4mm OD into the tap sitting perpendicular on the face of the vacuum connector.
6. Connect the vacuum cleaner to vacuum pump connector, using PVC piping as necessary to extend length as desired.
7. Power electronic sensor box using an appropriate power supply with the provided USB-B to USB-A cable.

## 4. End-of-Life Plan

### *4.1. Experimental Rig*

The experimental rig will be retained at Marshgate, UCL East and reused in the following academic year for MECH0085 MSc Individual Project, supervised by Prof. Castrejón-Pita, in collaboration with a global agricultural technology company. All PLA components are reprintable from the CAD files provided in the GitHub repository should replacement parts be required. Aluminium profile sections and acrylic panels should be stored flat and protected from impact damage.

### 4.2. Aerosol Extraction System Prototype

The aerosol extraction system prototype will be retained at Marshgate, UCL East for future student projects in collaboration with Agfa Inkjet Solutions. Transfer of prototype has been organised with Prof. Castrejón-Pita to be moved to the Manufacturing Futures Laboratory at Marshgate, UCL East after the 2026 Mechanical Engineering Summer Session. The OpenFOAM computational model is available on GitHub for use in future student projects, and comes with a separate handover document as part of the handover package.

### 4.3. Disposal of Materials

- PLA offcuts and failed prints are disposed of in the PLA recycling bin on the 5th Floor workshop inside MechSpace.
- Acrylic sheets that are unsuitable for reuse are disposed of in the appropriate recycling bins.

## 5. Key Contacts

For queries relating to the project, rig, or prototype, contact the relevant parties listed below.

Table 4: Key Project Contacts

| Role | Name | Contact |
| --- | --- | --- |
| Project Supervisor | Prof. José Rafael Castrejón-Pita | r.pita@ucl.ac.uk |
| Team Member | Teik Beng Goh | derrick.goh.22@alumni.ucl.ac.uk |
| Team Member | Nolan Lau Cze Kai | nolan.lau.22@alumni.ucl.ac.uk |
| Team Member | Tengku Amran Sharif | amran.sadruddin.22@alumni.ucl.ac.uk |
| Team Member | Yuxiang Zhou | yuxiang.zhou.22@alumni.ucl.ac.uk |

## 6. Declarations

This handover document has been prepared by the Aerosol Extraction System group as a component of the MECH0073 prototype handover package. The team confirms that:

- All physical items listed in Section 2 are present and accounted for in the designated storage space provided by Prof. José Rafael Castrejón-Pita at the Manufacturing Futures Laboratory at Marshgate, UCL East.
- All documentation listed in Section 2.3 has been submitted and is accessible via the project GitHub repository.
- All risk assessments are fully signed off and available in Appendix A of the Final Report, on the OSHENS risk management system, and the central GitHub repository accessible via QR code below.

Handover Repo

Figure 1: Handover Repository QR Code

Code Repo

Figure 2: Code Repository QR Code